Internship Final Report

Student Name: Aarav Raina

University: central University of jammu

Major: Computer Science

Internship Duration: JUNE 1st, 2026 - JUNE 30, 2026

Company: ShadowFox

Domain: AI/ML

Mentor: Mr. Hariharan

Coordinator: Mr. Aakash

Objectives

Develop and evaluate an image classification model using TensorFlow.

Tasks and Responsibilities

Got dataset of CIFAR-10, built CNN, trained model, evaluated accuracy.

Learning Outcomes

Learned CNN fundamentals, training, evaluation.

Challenges and Solutions

Resolved TensorFlow installation issues by using Google Colab because Python 3.14 is not yet supported. Improved model accuracy by debugging and tuning the model.

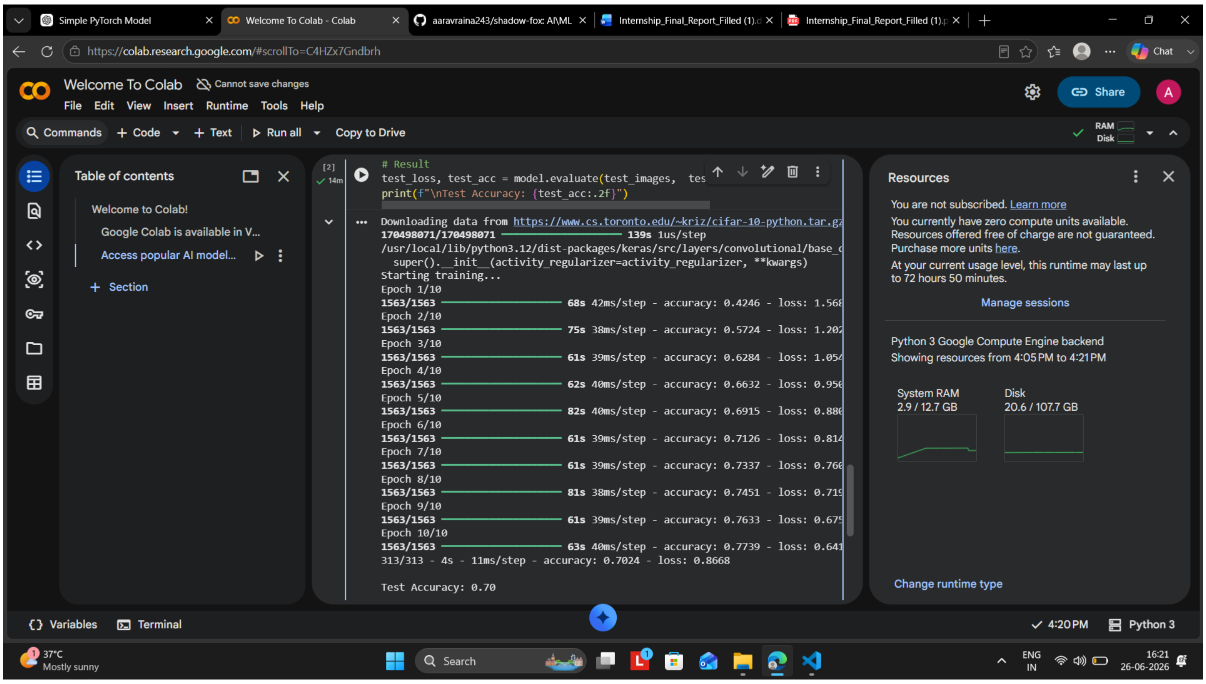
Conclusion

The project successfully demonstrated image classification using deep learning.

Acknowledgments

Thanks to ShadowFox, the mentor, for their guidance and material.

Training Result Screenshot


Simple PyTorch Model
Welcome To Colab - Colab
aaravraina243/shadow-fox: AI\ML
Internship_Final_Report_Filled (1).
https://colab.research.google.com/#scrollTo=C4HZx7Gndbrh
Welcome To Colab
Cannot save changes
File Edit View Insert Runtime Tools Help
Share
Commands
+ Code
+ Text
Run all
Copy to Drive
RAM
Disk
Table of contents
Welcome to Colab!
Google Colab is available in V...
Access popular AI model...
+ Section
[2]
14m
Result
test_loss, test_acc = model.evaluate(test_images, tes
print(f"\nTest Accuracy: {test_acc:.2f}")
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 139s 1us/step
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_c
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Starting training...
Epoch 1/10
1563/1563 68s 42ms/step - accuracy: 0.4246 - loss: 1.568
Epoch 2/10
1563/1563 75s 38ms/step - accuracy: 0.5724 - loss: 1.202
Epoch 3/10
1563/1563 61s 39ms/step - accuracy: 0.6284 - loss: 1.054
Epoch 4/10
1563/1563 62s 40ms/step - accuracy: 0.6632 - loss: 0.950
Epoch 5/10
1563/1563 82s 40ms/step - accuracy: 0.6915 - loss: 0.880
Epoch 6/10
1563/1563 61s 39ms/step - accuracy: 0.7126 - loss: 0.814
Epoch 7/10
1563/1563 61s 39ms/step - accuracy: 0.7337 - loss: 0.760
Epoch 8/10
1563/1563 81s 38ms/step - accuracy: 0.7451 - loss: 0.719
Epoch 9/10
1563/1563 61s 39ms/step - accuracy: 0.7633 - loss: 0.675
Epoch 10/10
1563/1563 63s 40ms/step - accuracy: 0.7739 - loss: 0.641
313/313 - 4s - 11ms/step - accuracy: 0.7024 - loss: 0.8668
Test Accuracy: 0.70
Resources
You are not subscribed. Learn more
You currently have zero compute units available. Resources offered free of charge are not guaranteed. Purchase more units here.
At your current usage level, this runtime may last up to 72 hours 50 minutes.
Manage sessions
Python 3 Google Compute Engine backend
Showing resources from 4:05 PM to 4:21 PM
System RAM
2.9 / 12.7 GB
Disk
20.6 / 107.7 GB
Change runtime type
Variables
Terminal
4:20 PM
Python 3
37°C
Mostly sunny
Search
ENG
IN
16:21
26-06-2026